

Section 4: Solution Design – SageMaker Inference Optimization

Problem Statement

The courses table contains 200 million records. A machine learning model hosted on an AWS SageMaker endpoint predicts skillsets from course descriptions. The current pipeline requires approximately two months to process all records.

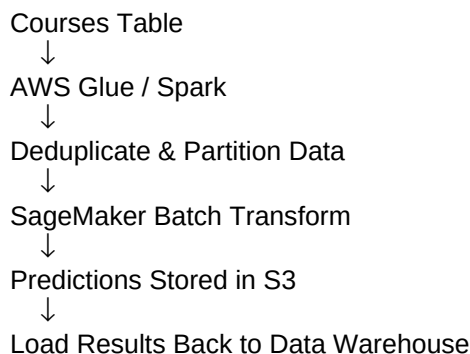
Current Bottleneck

The pipeline performs 200 million sequential API calls to the SageMaker endpoint. Each request incurs network latency, serialization overhead, endpoint processing time, and response handling, creating a significant throughput bottleneck.

Optimization Strategy

- Use AWS Glue / Apache Spark for distributed processing.
- Partition datasets across executors for parallel inference.
- Batch multiple course descriptions into a single request.
- Enable SageMaker endpoint auto-scaling.
- Use larger CPU/GPU-backed endpoint instances when appropriate.
- Replace real-time inference with SageMaker Batch Transform for large offline workloads.
- Cache predictions for duplicate descriptions.
- Process only incremental/new records instead of rescoreing historical data.

Recommended Architecture



Final Recommendation

The largest performance improvement comes from combining distributed Spark processing with SageMaker Batch Transform. Together with batching, auto-scaling, caching, and incremental processing, runtime can be reduced from months to hours while improving scalability and lowering operational costs.